

PROJECT HORIZON

The Device Lifecycle Intelligence System

Turning Equipment Obsolescence into a Retention Advantage

Turning predictable device churn into measurable contract renewals.

Executive Summary

Four answers. One page.

Device Cliff

THE PROBLEM

Customers with aging handsets churn at predictable thresholds - undetected today

Project Horizon

THE SOLUTION

Monthly ML scoring triggers device upgrade offers before customers defect

PR-AUC 0.685

THE MODEL

91% Recall · F1 0.689 · ValidationWeightedEnsemble

3× ROI (Base)

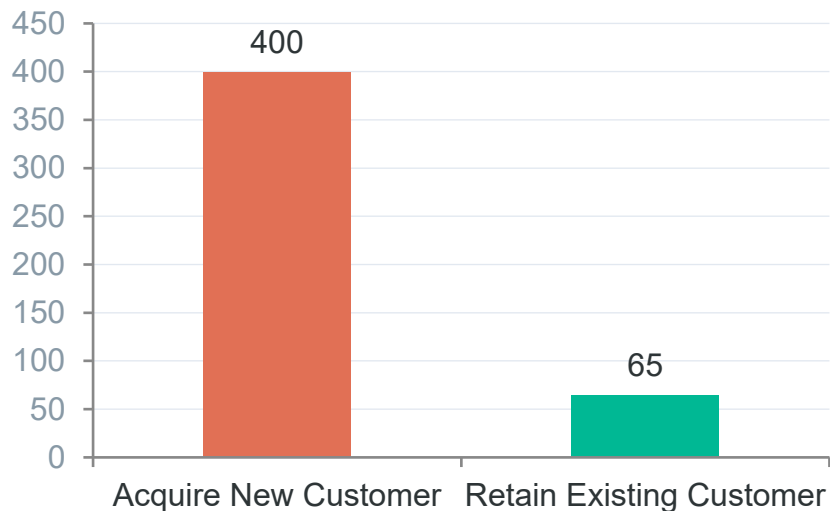
THE RETURN

18-month contract renewal per retained subscriber - targeting top decile

Market Context 2026 - Why Retention Is the Only Growth Lever

Saturated markets, rising acquisition costs, and the 5G device cycle converge.

Customer Acquisition vs Retention Cost



\$300–\$500

Customer Acquisition Cost per gross add

>100%

SIM penetration in mature US market - all growth is competitive displacement

5–8 mo

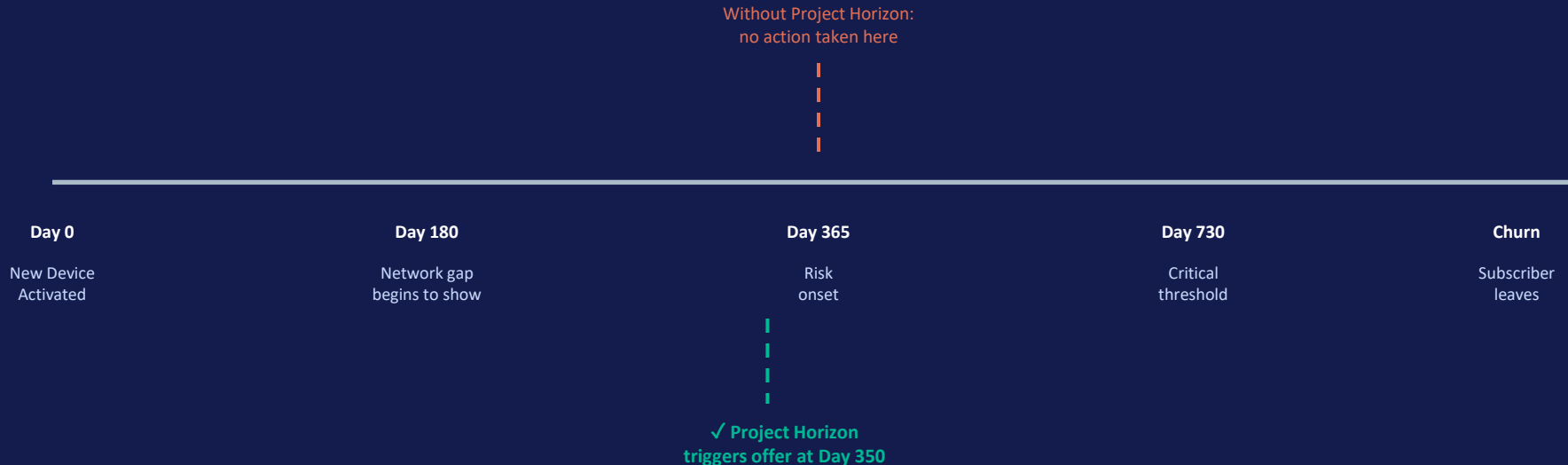
Break-even months just to recover the cost of winning one new subscriber

Retaining one customer costs 84% less than acquiring a new one.

Sources: GSMA Intelligence, The Mobile Economy 2025; Bain & Company Telecom Retention Economics 2024.

The Core Problem - Company A Is Losing Its Best Customers Silently

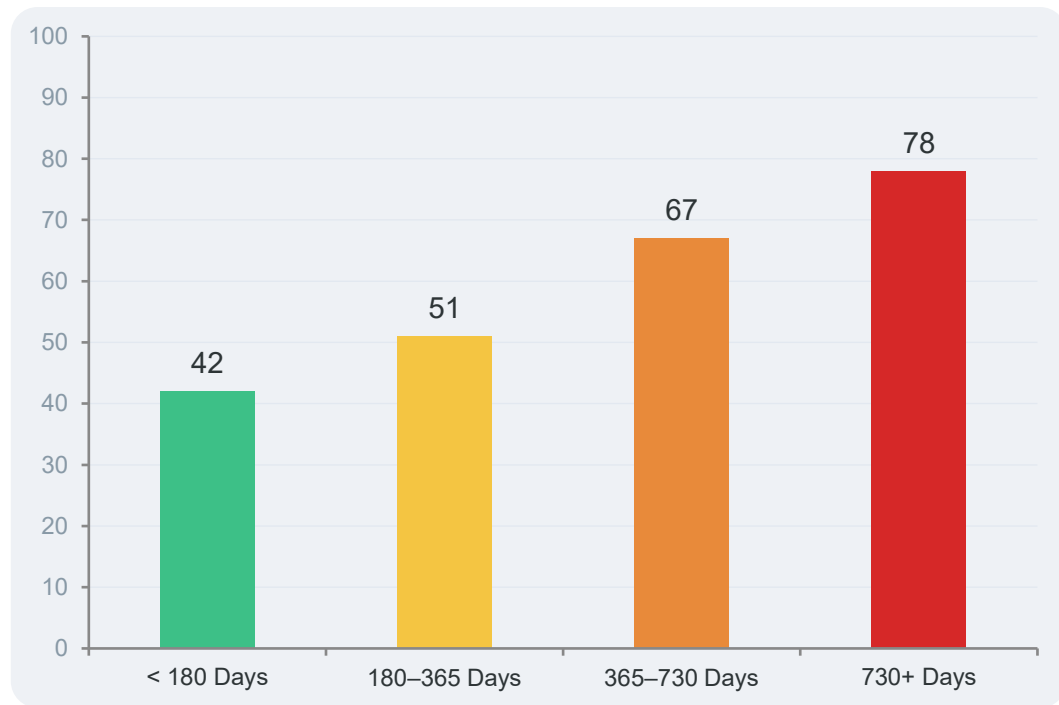
High-revenue subscribers defect for a reason the business has never measured.



Key Finding: Analysis indicates customers with older devices exhibit significantly higher churn rates - and device age is the only major churn driver that can be reset to zero with a \$65 upgrade offer.

The Hidden Driver - The Device Age Cliff

Churn does not drift upward gradually. It jumps at predictable lifecycle thresholds.



86%

Lift in churn risk from <180 days to 730+ days device cohort

$p < 0.001$

K-S test confirms eqpdays distribution diverges significantly between churners and non-churners

Day 350

Project Horizon upgrade trigger point - intercept customers before the cliff

Source: EDA - Churn Rate by Device Age Band (Notebook Figure 03). · K-S statistical validation

Three Compounding Risk Signals

The most dangerous customers simultaneously carry all three indicators.

SIGNAL 1 Device Age

eqpdays > 365 days

Handset crosses lifecycle threshold. Network capability gap widens. Customer experience degrades.

ACTION: Subsidized upgrade offer resets lifecycle to Day 0.

SIGNAL 2 Service Friction

custcare_Mean elevated

High support call frequency indicates accumulated frustration - often rooted in device-related issues.

ACTION: Service recovery program before cancellation intent forms.

SIGNAL 3 Revenue Value

rev_Mean above average

High-ARPU customers represent disproportionate revenue leakage when they churn.

ACTION: Priority Tier 1 targeting in every campaign wave.

Customers in all three zones = PRIORITY 1 - highest expected net value per intervention dollar

Data Architecture & Feature Engineering

Two raw files → 20+ intelligence signals that machines see and humans miss.



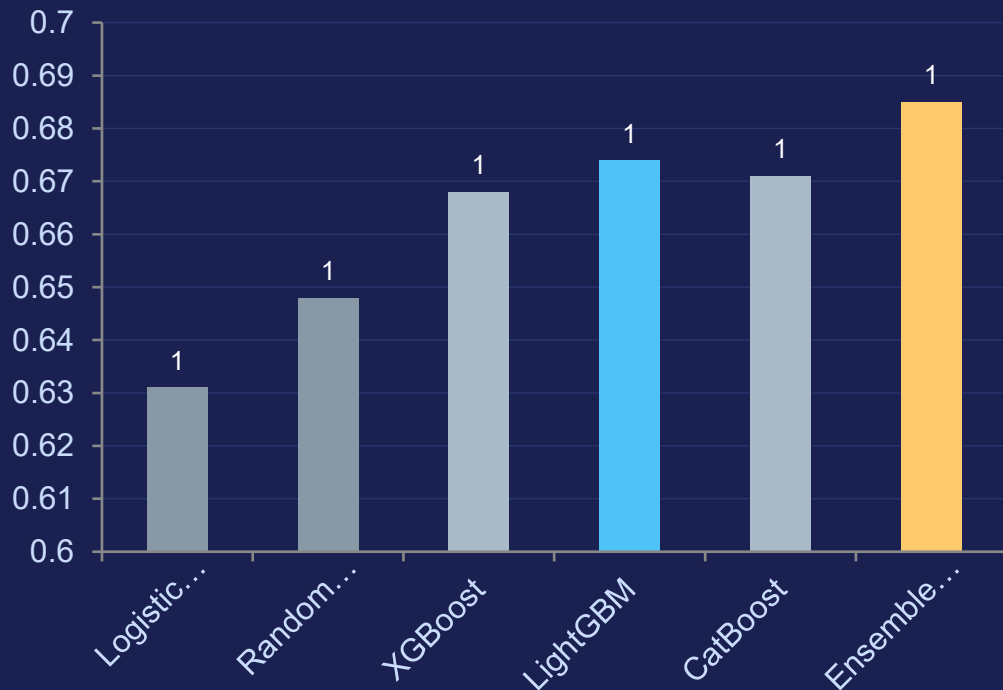
KEY ENGINEERED FEATURES

Feature	What It Measures	Business Meaning
Device Age Band	Lifecycle stage of handset	Identifies the 365-day danger zone
Custcare Intensity	Lifetime service friction score	Cumulative customer pain indicator
CLV Proxy	Revenue × Tenure	Quantifies what is at stake if customer leaves
Revenue Efficiency	Revenue per minute of use	Identifies over-paying, under-using accounts
Multi-Line Flag	Household vs. individual account	Structural switching cost - family plans churn less
Customer Value Score	Percentile composite ranking	Unified retention priority index
Device Age × Custcare	Interaction: old device + support calls	Compound risk - the highest-risk combination
KMeans + GMM Segments	5 behavioral clusters (train-fitted)	Group-level targeting - no leakage into holdout

All train-derived features (percentile ranks, cluster labels) are fitted on training data only - preventing any leakage into validation or holdout sets.

The Intelligence Engine - Model Selection & Validation

Five algorithms benchmarked. One winner selected by evidence, not preference.



5-Fold CV

Stratified cross-validation for all 5 model families to prevent overfitting

50 Trials

Optuna TPE hyperparameter optimization on LightGBM to find global optimum

80/15/5%

ValidationWeightedEnsemble blend: OptunaLightGBM · LightGBM · CatBoost

Why an Ensemble?

Single models captured different churn patterns. Combining Optuna LightGBM, LightGBM, and CatBoost improved PR-AUC from 0.674 (best individual model) to 0.685 (+1.6%), producing the strongest ranking of high-risk customers

Why PR-AUC - not Accuracy? A model that always predicts 'No Churn' is technically accurate but commercially useless. PR-AUC rewards finding the rare, high-value churner.

Model Performance - The Mandatory Evidence Slide

Champion: ValidationWeightedEnsemble (OptunaLightGBM 80% · LightGBM 15% · CatBoost 5%)

91%

RECALL

For every 100 customers who will cancel, our system identifies 91 before they act.

Metric	Score	Business Meaning
PR-AUC (primary)	0.685	Ranking quality across all decision thresholds
ROC-AUC	0.696	Overall discrimination between churners and non-churners
F1-Score	0.689	Balanced precision-recall for operational targeting
Precision	55.5%	55% of flagged customers will actually churn
Recall	91.0%	91% of actual churners are caught before defection

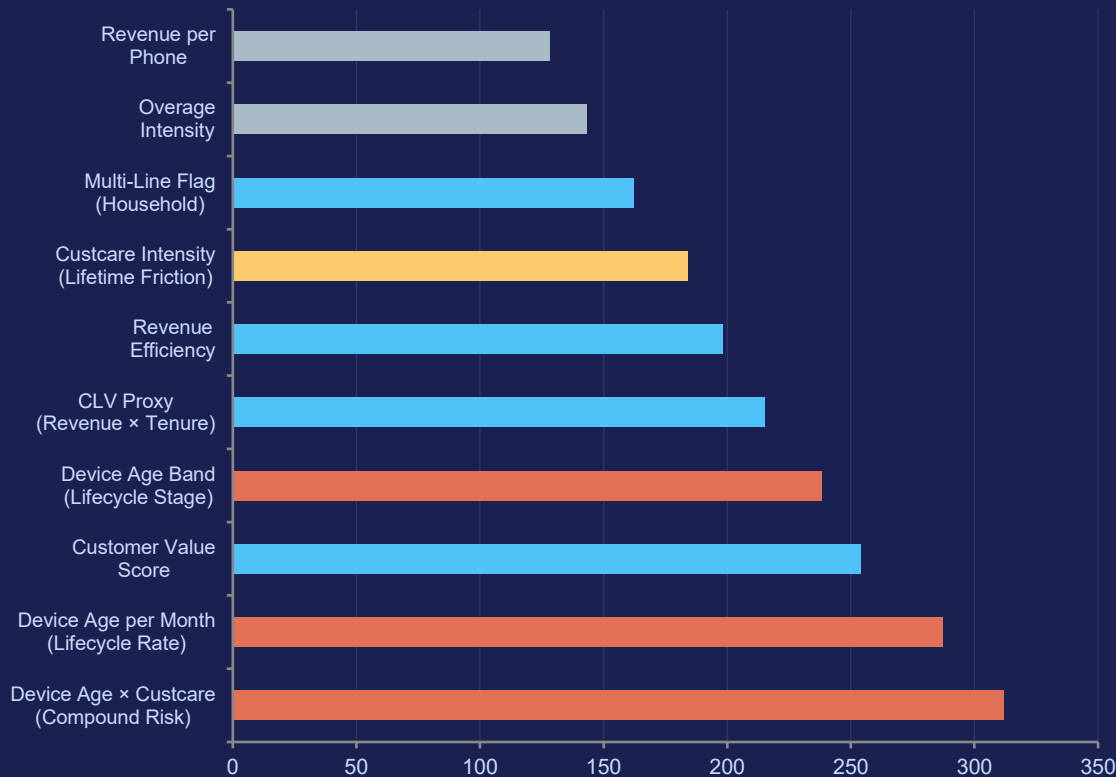
The Business Logic Behind 91% Recall

Missing a churner (False Negative): \$55/month × 18 months = **\$990 in lost revenue**

Offering to a non-churner (False Positive): \$65 in campaign cost - a 15× cost asymmetry. 91% Recall is the correct business calibration.

What Drives the Risk? - The Operational Signal Map

Machine confirms hypothesis: device age and service friction dominate churn prediction.



Source: Optuna LightGBM Feature Importance Analysis (Figure 19).

Directly actionable - device upgrade resets these

Revenue & behavioral signals

Service friction indicators

Structural / demographic signals

Lifecycle Signals

Device Age x Custcare
Device Age per Month
Device Age Band

Customer Value Signals

Customer Value Score
CLV Proxy

Service Friction Signals

Custcare Intensity

Account Structure Signals

Multi-Line Flag

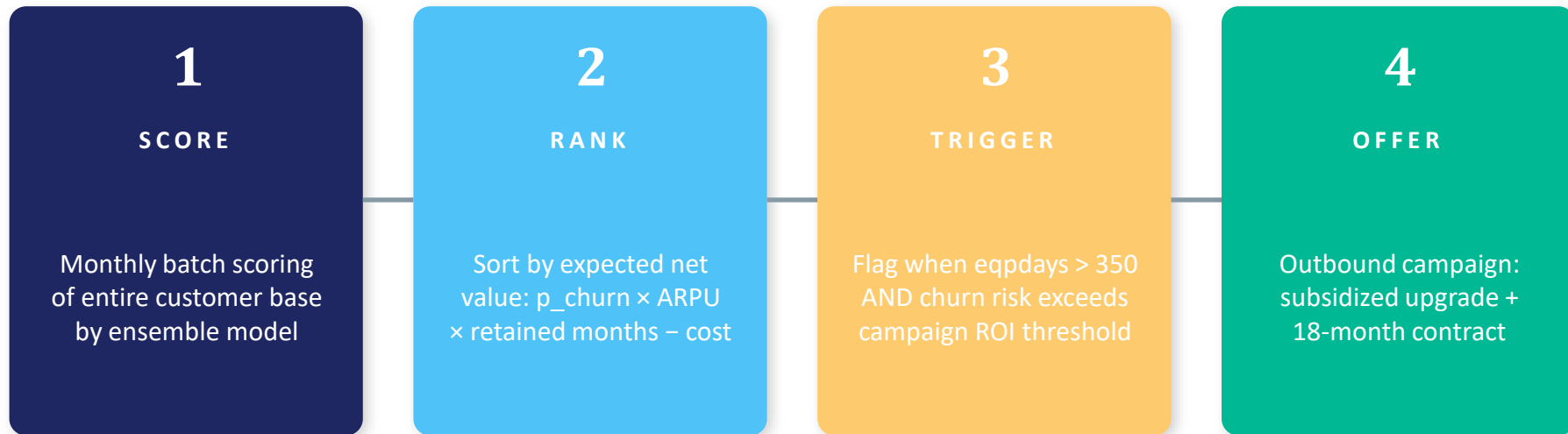
Machine confirms that lifecycle and service-friction signals dominate churn prediction.

Equipment age ranks in the top predictors. Unlike demographics or tenure - which cannot be changed - device age can be reset to zero with a subsidized upgrade.

This closes the evidence loop:
Hypothesis → Data → Model → Campaign.

The Business Proposal - Project Horizon

A monthly automated loop that converts device age into contract renewals.



TARGETING MATRIX

	LOW CHURN RISK	HIGH CHURN RISK
HIGH REVENUE	MONITOR - Loyalty offer if risk increases	★ INTERVENE NOW Device upgrade + contract renewal
LOW REVENUE	MAINTAIN - Standard plan	SERVICE RECOVERY Friction reduction only

Top 10% by expected net value are activated for the first campaign wave each month.

Intervention Strategy - How the Model Becomes a Campaign

The scoring output is not a report. It is an automated trigger list with a specific action per customer.

CAMPAIGN TRIGGER LOGIC

```
IF eqpdays > 350
AND p_churn > calibrated threshold
AND customer ∈ top 10% by expected net value
THEN → Send subsidized device upgrade offer
```

COST OF BEING WRONG

False Positive
(over-targeted)
\$65 wasted per customer

False Negative
(missed churner)
\$990 in lost revenue
($\$55 \times 18$ months)

BASE OFFER DESIGN (per customer)

Handset subsidy	\$50 device credit
Marketing outreach	\$15 per contact
Total campaign cost	\$65
Contract term secured	18 months
Acceptance rate (Base Scenario)	30% (planning assumption)

15× Cost Asymmetry

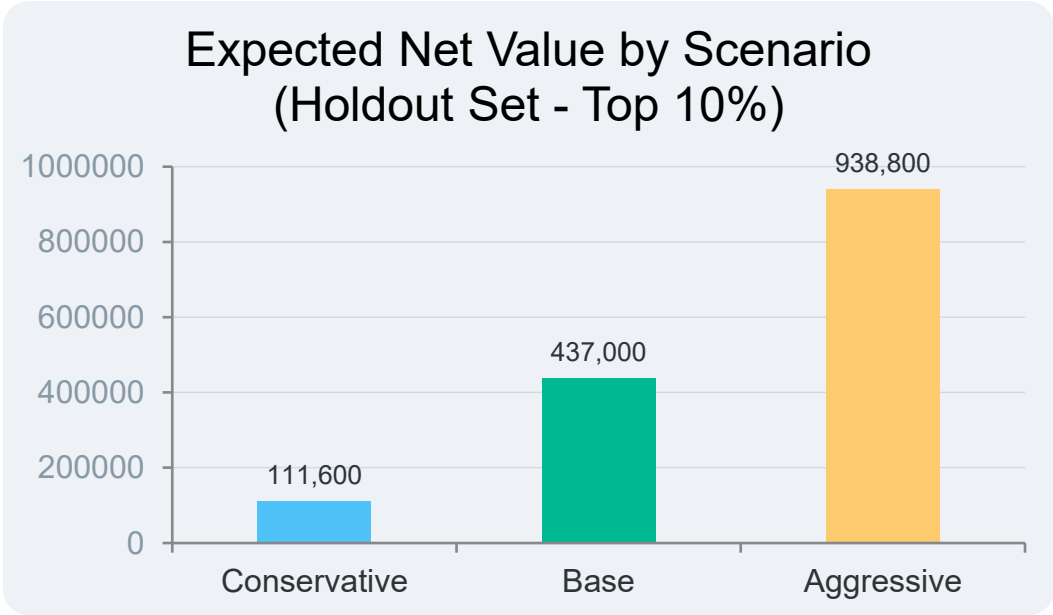
Missing one churner costs 15× more than a wasted upgrade offer. 91% Recall is the deliberate, justified business calibration - not a modeling accident.

Acceptance rates are scenario assumptions used for ROI modeling and are not derived from Company A data. Conservative (20%), Base (30%), and Aggressive (40%) scenarios are shown to illustrate sensitivity.

Scenario assumptions used for sensitivity analysis; not directly observed in Company A data.

Financial Impact - The ROI of Project Horizon

Three scenarios. Explicit assumptions. Defensible range - not a single optimistic number.



Scenario	Accept Rate	Retained Mo.	Cost/Customer	ROI
Conservative	20%	12 mo.	\$45	~1.2×
Base ★	30%	18 mo.	\$65	~3.4×
Aggressive	40%	24 mo.	\$85	~5.5×

Category	Source
Monthly Revenue (rev_Mean)	Observed in Company A dataset
Churn Probability (p_churn)	Predicted by ensemble model
Top-Decile Targeting	Derived from model ranking
Acceptance Rate	Scenario assumption (20/30/40%)
Retained Months	Scenario assumption (12/18/24)
Campaign Cost	Business assumption (\$45/\$65/\$85)

What Comes From Data?

✓ Revenue ✓ Churn probability ✓ Customer ranking

What Comes From Business Assumptions?

✓ Acceptance rate ✓ Contract duration ✓ Subsidy cost

Even at the Conservative scenario (20% acceptance, 12-month retention, \$45 cost), every targeted customer with ARPU above ~\$38 produces a positive expected return. US telecom ARPU of \$50–\$65 places virtually all targets above breakeven.

Note: Dollar figures are modeled estimates based on notebook formula applied to holdout set characteristics. Exact values are generated when notebook runs against full dataset. Revenue values derived from rev_Mean; ROI estimates generated using notebook expected-value framework.

Risks, Safeguards & How the Model Minimizes Them

Anticipating failure modes is what separates a consulting engagement from a student exercise.

CANNIBALIZATION

Upgrade offers given to customers who would renew anyway inflate cost.

MITIGATION:

Expected net value formula naturally filters low-probability churners. 55.5% precision is deliberate - not a flaw.

OFFER FATIGUE

Customers receiving excessive promotions disengage and acceptance rates fall.

MITIGATION:

90-day suppression window prevents over-contact. Monthly scoring cadence respects subscriber attention.

MODEL DRIFT

As 5G/6G device cycles accelerate, churn patterns may shift away from training distribution.

MITIGATION:

Quarterly model refresh with rolling 12-month churn labels. PR-AUC monitored against 0.682 baseline.

DATA PIPELINE

Delayed or miscoded eqpdays data from new device activations breaks trigger logic.

MITIGATION:

Automated quality alerts when eqpdays null rates exceed training baseline. Median imputation as fallback.

All four risks are anticipated and mitigated. The model's pipeline design (train/val/test split, ID drop, train-fitted encoders) eliminates the most common failure modes by construction.

References & Citations

All external claims, algorithms, datasets, and tools properly attributed.

ALGORITHMS & LIBRARIES

- [1] Ke et al. (2017). LightGBM. NeurIPS. <https://papers.nips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- [2] Chen & Guestrin (2016). XGBoost. KDD. <https://doi.org/10.1145/2939672.2939785>
- [3] Prokhorenkova et al. (2018). CatBoost. NeurIPS. <https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>
- [4] Akiba et al. (2019). Optuna. KDD. <https://doi.org/10.1145/3292500.3330701>
- [5] Pedregosa et al. (2011). Scikit-learn. JMLR, 12, 2825–2830.
- [6] Lundberg & Lee (2017). SHAP. NeurIPS. <https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [7] Harris et al. (2020). NumPy. Nature, 585, 357–362.
- [8] McKinney (2010). Pandas. SciPy Conf.

MARKET ANALYSIS

- [9] GSMA Intelligence (2025). The Mobile Economy 2025. <https://www.gsma.com/>
- [10] Bain & Company (2024). Telecom Retention Economics. <https://www.bain.com/>

DATA & TOOLS

- [11] GCI World (2026). Company A Dataset: Client.csv & Record.csv. Omnicampus Portal.
- [12] Anthropic (2026). Claude AI - Strategic Analysis Assistance. <https://claude.ai/>